

David Cain | <https://davidcain.uk>

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Graduate Computer Science Student from the University of Hull with First-Class Honours.

1 year of industry experience as a test automation engineer at an embedded electronics design company.

I have strong skills in Robot Framework, Python, C, embedded development, parallel programming, web and UI programming, C#, and full stack technologies.

Education:

BSc Computer Science with Industrial Experience – University of Hull

September 2021 – July 2025

- Grade: First-Class Honours
- Distributed Systems, Parallel and Concurrent Programming, Machine Learning, UI/UX, Placement Year

Year In Industry:

Automation Engineer & more – (Year in Industry) – Sabre Technology

July 2023 – July 2024

- Automated large-scale embedded system testing using Robot Framework to reduce manual test effort for team
- Collaborated across firmware/test teams in Git (tickets, pull requests, branching), and management in weekly meetings
- Developed internal tools to help team productivity:
 - Python LoRaWAN packet decoder library using YAML message schemas for extensibility
 - WPF frontend for a barcoding tool
 - Google Sheets Hex Message encoder and decoder

Selected Projects:

- **Cryptography & Authentication API – Uni Assignment – (C#, ASP.NET Web API, Entity Framework)**
 - REST endpoints for SHA hashing and signing, RSA and AES encryption, accounts, API keys, and more
 - Implemented in ASP.NET Web API, and Entity Framework with C# client using the API for local services
- **OV7670 Camera & Web Interface – Own Project – (C, CMake, JS, HTML, CSS, Python, Flask, KiCad)**
 - C RP2040 firmware and custom high frequency PCB to stream image data from bare-bones CMOS sensor module to Python Flask server over serial under microcontroller memory and clock limitations.
 - JS frontend uses the API to take images and set registers, rendering with HTML Canvas
- **Honours Stage Project – Uni Module – (JS, HTML, CSS, HTML Canvas)**
 - Created a zero-footprint web application for creating visual radiology reports for 3D medical scans.
 - Created a JavaScript rendering library for multiplanar reconstruction, slicing, segmentation, and rendering with a Web App interface and conducted user testing and evaluation
- **Parallel Programming – Uni Module – (CUDA, RUST, OpenGL)**
 - Utilised Cuda and multithreaded Rust to parallelise ray casting, bird flocking simulation, matrix multiplication, and more with massive performance boosts using memory tiling, locks, and more
- **Machine Learning – Uni Module – (TensorFlow, PyTorch, GAN, CNN)**
 - Created, trained, fine-tuned, and regularised deep CNNs and GANs in TensorFlow and PyTorch to solve written digit labelling and synthetic data generation. Accelerated with Google Colab cloud TPUs and GPUs
- **Full Portfolio:** <https://davidcain.uk>

Skills:

- C# • HTML, CSS, JavaScript • Robot Framework • C, C++ & CMake • Python • Cuda • Rust • SQL
- Git • AI, TensorFlow & PyTorch • Unity • Blender • KiCad • Figma • Google Colab
- Google Sheets + Excel • Matplotlib • Microsoft Azure
- Embedded • Agile • Waterfall • UML • Digital Art and Graphic Design • Video Editing • Electronics • WordPress

Achievements:

- **Best Certificate Academic Performance in Computer Science (University of Hull)** *2021 – 2022*
 - **AZ-900 Azure Cloud Fundamentals (Microsoft)** *May 2022*
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Additional Work Experience:

Email Marketing & Lead Acquisition – Perimeter Site Security *October 2025 – Present*

- Gather and filter construction leads from multiple channels for use across sales team
- Send personalised outreach emails to potential clients pitching site security
- Created web scraper scripts to gather contacts from online directories and websites

B2C Telesales Advisor – Clearanswer *July 2025 – October 2025*

- Utilised internal systems to sell, and manage customer accounts and contracts for BT and EE broadband Mobile and TV
- Collaborated with other departments to solve non sales needs
- Communicated technical details of sims, devices, and broadband technologies to consumers in clear and understandable manner to increase sales

Student Demonstrator – University of Hull *September 2022 – July 2023*

- Helped teach C# Object Oriented Programming to 1st year students in Labs
- Developed listening and explanatory skills by communicating and breaking down concepts to a variety of students in different modules

Part-Time Work – Various *2019, 2020, 2021*

- Work pre-university at various cafes, restaurants and fast paced environments